

Humphrey Afobhokhan

Healthcare AI Researcher & Data Scientist

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SUMMARY

Healthcare AI researcher and data scientist building predictive analytics systems, AI-driven healthcare products, and scalable clinical data workflows. Interdisciplinary background spanning microbiology, applied economics, and business analytics, with hands-on experience in Python, SQL, machine learning, and product analytics. Focused on patient-centered AI systems that improve clinical insight, longitudinal context, and healthcare accessibility.

EXPERIENCE

Data Scientist — Applied AI & Product Analytics — Medyl San Francisco, CA | Jan 2025 – Present

- Designed and shipped a patient-centered healthcare AI platform integrating structured and unstructured medical data to surface longitudinal patient context and AI-assisted health signals in the 48 hours before a clinical appointment.
- Built engagement analytics and product instrumentation across patient, clinician, and admin surfaces (onboarding funnels, symptom-reporting patterns, retention), informing iteration toward a pilot-ready behavioral product.
- Architected scalable healthcare data infrastructure on Firebase/Firestore and Next.js prioritizing portability, interoperability, and patient-controlled access, deployed across three production surfaces.
- Authored three peer-reviewed SSRN papers on patient-owned health data infrastructure, AI governance, and interoperable care, establishing a research foundation for the underlying data architecture.

Researcher — Computer Vision for Clinical Microscopy — MicroSight AI San Francisco, CA | Jul 2025 – Present

- Designing an AI-powered computer vision system for real-time microscopy-assisted diagnostic support, targeting clinical workflows in resource-constrained settings.
- Investigating scalable image-processing and intelligent-detection methods for clinical imaging, including model selection, dataset curation, and deployment considerations.

Database Coordinator — Analytics & ETL — Kayode Trading Dubai, UAE | Sep 2019 – Jul 2022

- Boosted operational throughput by 22% using predictive analytics, SQL pipelines, and Power BI dashboards across procurement and inventory operations.
- Integrated Snowflake and automated ETL pipelines, standardizing cross-functional reporting and reducing manual data-prep time across teams.

Biomedical Analyst — Nigerian National Petroleum Corporation Medical Services Benin, Nigeria | Jan 2017 – Jun 2019

- Conducted laboratory diagnostics and optimized data accuracy across microbiology and hematology units, applying biostatistical methods to sample analysis and quality control.

SELECTED PUBLICATIONS

- “A Patient-Owned Health Data Layer for AI-Enabled Healthcare Systems.” SSRN, 2026
- “Patient-Centered Health Data Architectures for Interoperable Care in Emerging and Underserved Markets.” SSRN, 2026
- “Governing Patient-Centered Health Data Systems: Trust, Consent, and AI Accountability in Emerging Digital Health Ecosystems.” SSRN, 2026

SELECTED PROJECTS

Clinical NLP Triage Classifier Python, HuggingFace Transformers, PyTorch, Streamlit, Docker | 2026

Live demo: huggingface.co/spaces/Osafobz/clinical-triage-classifier · Repo: github.com/Cyberpunk379/clinical-triage-classifier

- Built a zero-shot triage classifier and symptom extractor for free-text clinical notes, deployed as a public Streamlit demo on HuggingFace Spaces with a Dockerized BART-MNLI inference pipeline.
- Implemented NegEx-style negation handling that correctly suppresses lexicon matches inside scope (e.g., "no chest pain"), differentiating the pipeline from naive keyword-based clinical extractors.
- Achieved 100% accuracy (6/6) on a hand-labeled evaluation set spanning all three triage tiers, with a reproducible eval harness and 15 unit tests covering the negation logic and rationale rendering.
- Designed model-agnostic architecture with a live UI selector between a 300MB distilled BART-MNLI and 1.6GB BART-large-MNLI, demonstrating cost/accuracy tradeoffs in production NLP systems.

EDUCATION

M.S., Applied Economics (in progress) — University of San Francisco San Francisco, CA | Expected May 2027

Coursework: Econometrics, Computation for Data Science, Probability & Statistics, Mathematics for Economists, Machine Learning

Doctor of Business Administration (in progress) — Hult International Business School San Francisco, CA | In Progress

Research focus: Predictive Health Analytics and Data Infrastructure

MBA, M.S. Entrepreneurship & Innovation, M.S. Business Analytics — Hult International Business School

London, UK & San Francisco, CA | Apr 2024

President, Hult Data Global. Coursework: Python, SQL, Machine Learning, AI, Data Visualization, Finance, Strategy.

B.S., Microbiology — Benson Idahosa University Benin, Nigeria | Apr 2018

Focus: Biostatistics, Biotechnology

TECHNICAL SKILLS

Languages & ML: Python, R, SQL, PyTorch, HuggingFace Transformers, scikit-learn, NLP, Zero-shot Classification, Computer Vision, Statistical Modeling, Predictive Analytics

Data & Infrastructure: Snowflake, Databricks, dbt, ETL pipelines, MySQL, NoSQL, Firebase/Firestore, Docker

Product & Visualization: Streamlit, Tableau, Power BI, Next.js, Product Analytics, HuggingFace Spaces

LANGUAGES

English (fluent), Yoruba (fluent), Esan (native), French (intermediate), Georgian (intermediate)